WhoSThis: Real-Time Speaker Identification via Multi-Teacher Distillation

September 10, 2026

Davide Pietragalla

Abstract

Recent speaker verification models have demonstrated remarkable robustness and consistency across major datasets. In this report, we present a multi-teacher distillation process for speaker verification that synthesizes recent state-of-the-art advancements. This approach yields a highly efficient model capable of real-time speaker verification. To demonstrate its practical effectiveness, we integrate the model into a comprehensive system featuring an intuitive user interface.

1. Introduction

Suppose you want to build a video calling system capable of diarizing conversations and identifying speakers in real time. To achieve this, the underlying speaker identification component must be exceptionally fast and efficient. The objective of this work is to employ a teacher distillation approach to minimize latency while sufficiently preserving qualitative accuracy. Finally, we embed this optimized model into a live system to show its ability to recognize active speakers in a room instantaneously.

2. Related Works

The Time Delay Neural Network (TDNN) (Snyder et al., 2018) serves as the foundational architecture for both the student model and one of the two teachers.

ECAPA-TDNN, one of the two teachers, improves upon the base architecture by restructuring the frame layers into Res2Net-style modules (Gao et al., 2021) enhanced with Squeeze-and-Excitation channel attention (Hu et al., 2019).

A parallel line of work from NVIDIA focused on efficient 1D depthwise separable convolutional encoders, culminating in TitaNet (Koluguri et al., 2021), the second teacher.

A pooling mechanism is essential for converting a variable-length sequence of frame-level features into a fixed-size utterance embedding. Okabe et al. (Okabe et al., 2018) extended traditional statistics pooling into Attentive Statistics Pooling (ASP).

Multi-scale aggregation before pooling is proven to boost model performance, as demonstrated by MFA-Conformer (Zhang et al., 2022), which fuses local and global representations to improve accuracy and inference speed.

Xiang et al. (Xiang et al., 2019) adapted Additive Angular Margin (AAM) (Wang et al., 2018) — commonly referred to as AAM-Softmax — as the standard training objective for TDNN.

The lightweight model is of course trained using knowledge distillation, a foundational framework introduced by Hinton et al. (Hinton et al., 2015).

3. Methodology

This work proposes a multi-teacher knowledge distillation process that compresses two state-of-the-art speaker verification models — SpeechBrain's ECAPA-TDNN (Desplanques et al., 2020) and NVIDIA NeMo's TitaNet-Large (Koluguri et al., 2021) — into a single lightweight student network. The student shares ECAPA-TDNN's macro-architecture at half capacity, but is trained under a composite objective that combines supervised speaker classification with three simultaneous distillation signals.

3.1. Datasets

- VoxCeleb (Nagrani et al., 2017; Chung et al., 2018), a large-scale audio-visual speaker corpus scraped from YouTube interviews, provides both the training and in-domain evaluation data.

- LibriSpeech (Panayotov et al., 2015), a corpus of public-domain audiobooks, serves exclusively as an out-of-domain test set. Excluded from training, it tests how well embeddings learned on "in the wild" VoxCeleb conversations generalize to cleaner, non-conversational acoustics.

- MUSAN (Snyder et al., 2015) is a corpus of music,

Email: Davide Pietragalla <pietragalla.1965270@studenti.uniroma1.it>.

Deep Learning and Applied AI 2026, Sapienza University of Rome, 2nd semester a.y. 2025/2026.

speech, and noise recordings and is used only at evaluation time to build a controlled, noise-stressed test condition.

Two SpecAugment-style masks (Park et al., 2019) are stochastically applied to the transposed time-frequency representation exclusively during training: frequency masking (up to 15 mel-bins) and time masking (up to 50 frames).

To enable uniform batching, training utterances are cropped or zero-padded to a fixed 3-second window. Conversely, evaluation utterances retain their natural lengths up to a 6-second cap, while any utterance under three seconds is discarded to ensure informative embeddings.

3.2. Student Architecture

The student follows the same macro-architecture as its ECAPA-TDNN teacher, but at reduced capacity and with some structural modifications.

The encoder's default single-scale pooling is replaced by a module operating at multiple temporal granularities. For scales $n \in \{1, 2, 4\}$, the feature map is segmented over time into n contiguous chunks. Following Okabe et al. (Okabe et al., 2018), attentive statistics pooling (ASP) is applied within each chunk using its masked mean and standard deviation to compute attention weights. The resulting statistics are averaged across the n chunks for each scale. These per-scale pairs are concatenated and fused via a grouped 1×1 convolutional bottleneck (Conv1d → BatchNorm1d → GELU → Conv1d), reducing the $2 \cdot C \cdot |scales|$ channels to $2C$. Finally, a linear layer projects this fused representation into the 192-dimensional student embedding.

Three lightweight heads project the shared 192-dimensional student embedding into teacher-aligned spaces (Romero et al., 2015) without altering the base representation used for classification:

- `head_ecapa` and `head_titanet`: Two-layer MLPs (Linear → BatchNorm1d → GELU → Linear) that project the student core into 192-dimensional spaces aligned with the ECAPA and TitaNet teacher embeddings, respectively.

- `head_ecapa_feat`: A two-layer 1D convolutional head (Conv1d → BatchNorm1d → GELU → Conv1d) that operates on the student's unpooled MFA intermediate feature map to temporally average-pool these features and project them to match the teacher's intermediate dimensionality.

3.3. Training

During training, the student core embedding is passed to an Additive Angular Margin Softmax (AAM-Softmax) classification layer with scale=30 and margin=0.20. Concurrently, each of the three distillation losses is computed as:

$$\mathcal{L}_{\text{dist}} = 1 - \frac{1}{N}\sum_{i=1}^{N}\cos(s_i, t_i)$$

where N is the batch size (128), and $\cos(s_i, t_i)$ denotes the cosine similarity between the student projection s_i and its corresponding precomputed teacher target t_i.

To balance these objectives, the four loss components are dynamically weighted throughout the training process. This schedule allows early stages to prioritize teacher distillation, while later stages shift focus toward sharpening class discrimination.

The network is optimized using AdamW (learning rate $= 10^{-3}$, weight decay $= 10^{-4}$) paired with a CosineAnnealingLR schedule ($T_{max} = 150$ epochs, $\eta_{min} = 10^{-6}$). Finally, automatic mixed precision (FP16 autocast coupled with a GradScaler) is employed to maximize compute and memory efficiency.

4. Evaluation & Results

The evaluation uses a subset of VoxCeleb and the entire LibriSpeech dataset, measuring parameters, inference time per sample, EER (%), and MinDCF. Since LibriSpeech is a clean, noise-free dataset, we augmented it with dynamic noise using MUSAN. Another key difference between the two is that LibriSpeech contains scripted speech rather than natural dialogues. The results are the following.

Data	Model	Params	Time/Samp	EER	mDCF
VC	ECAP	22.15M	5.43ms	0.55%	0.0520
	TITA	25.33M	4.57ms	0.39%	0.0281
	STUD	7.66M	2.21ms	2.35%	0.2054
LS	ECAP	22.15M	5.37ms	1.75%	0.1051
	TITA	25.33M	4.62ms	0.86%	0.0637
	STUD	7.66M	2.22ms	4.50%	0.3808

5. Conclusions

As demonstrated in the table, the number of parameters was reduced to a third, roughly halving the inference time while maintaining acceptable EER and mDCF levels. Furthermore, the model has been integrated into the system WhoSThis and is fully operational for speaker identification.

Future research directions will focus not only on advanced process optimization and the integration of unseen state-of-the-art distillation techniques, but also on the approaches outlined in the next additional section describing the WhoSThis system.

6. Appendix: The WhoSThis System

WhoSThis is an application that leverages the student model for real-time speaker identification. An overview is shown in Figure 1.

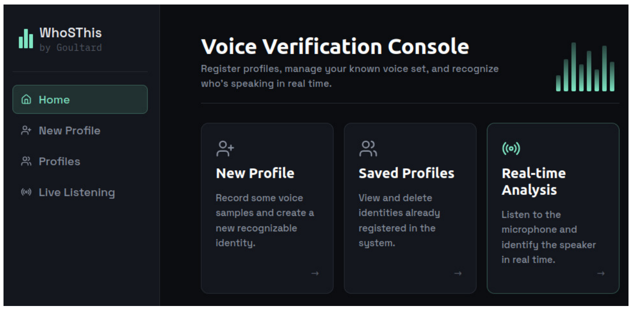

Figure 1. Home section of the system.

The system features three core functionalities:

- Profile Creation: As shown in Figure 2, users can record their voice multiple times. The system generates a unified speaker profile by averaging the embeddings extracted from each audio sample.

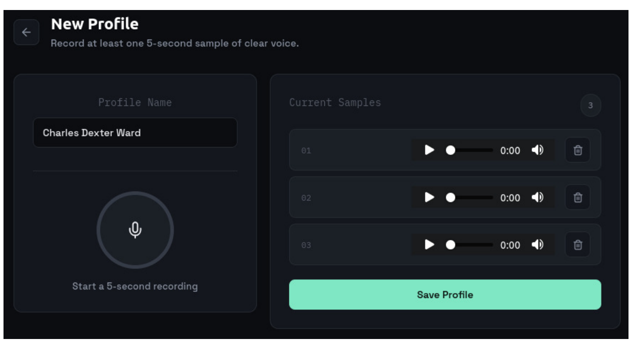

Figure 2. New Profile section of the system.

- Profile Modification: In Figure 3, it is possible to see that currently, users can delete their existing profile from the database to create a new, more accurate one. A planned future enhancement will allow users to delete specific low-quality audio samples without restarting the entire enrollment process.

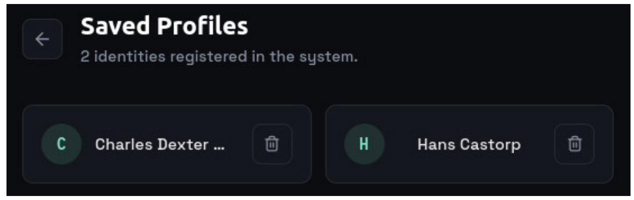

Figure 3. Saved Profiles section of the system.

- Live Listening: The device monitors a room in real-time and compares detected speech against the database. Using a threshold computed during evaluation, it identifies whether a registered user is speaking, an unknown person is talking, or if there is silence. The UI is shown in Figure 4.

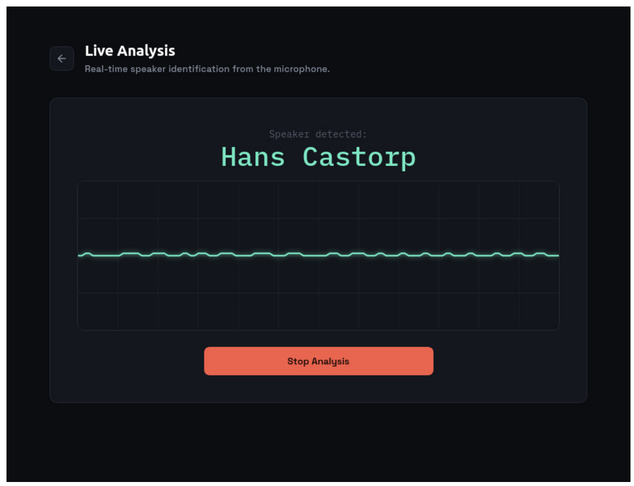

Figure 4. Real-Time Analysis section of the system.

Future improvements could involve fine-tuning the model directly on the stored audio database, or using reinforcement learning from human feedback (RLHF) to continuously adapt and improve accuracy.

7. Statement on the use of AI

We initially used LLMs to gather information, brainstorm, and outline a solid and comprehensive research of the topic.

Throughout the writing of this report, I relied on LLMs and tools like Grammarly to rephrase text and expedite proofreading and error correction.

Additionally, LLMs were employed to resolve library dependencies and troubleshoot the numerous errors that occurred during the early phases of development.

LLMs were instrumental in designing a training pipeline to manage OOM errors and speed up the entire process. Because OOM crashes still occurred sometimes due to Kaggle limits, LLMs also assisted in implementing a checkpointing logic to safely resume training and minimize lost time.

I heavily relied on LLMs for the design of the view of WhoSThis after the initial prototypes were created, primarily due to time constraints.

References

Chung, J. S., Nagrani, A., and Zisserman, A. Voxceleb2: Deep speaker recognition. In *Interspeech 2018*, interspeech$_2$018, $pp.$ 1086–1090.$ISCA$, 2018.doi : . URL http://dx.doi.org/10.21437/Interspeech.2018-1929.

Desplanques, B., Thienpondt, J., and Demuynck, K. Ecapa-tdnn: Emphasized channel attention, propagation and aggregation in tdnn based speaker verification. In *Interspeech 2020*, interspeech$_2$020, $pp.$ 3830–3834.$ISCA$, $October$2020.10.21437/$interspeech$.2020 − 2650.URL.

Gao, S.-H., Cheng, M.-M., Zhao, K., Zhang, X.-Y., Yang, M.-H., and Torr, P. Res2net: A new multi-scale backbone architecture. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 43(2):652–662, February 2021. ISSN 1939-3539. 10.1109/tpami.2019.2938758. URL http://dx.doi.org/10.1109/TPAMI.2019.2938758.

Hinton, G., Vinyals, O., and Dean, J. Distilling the knowledge in a neural network, 2015. URL https://arxiv.org/abs/1503.02531.

Hu, J., Shen, L., Albanie, S., Sun, G., and Wu, E. Squeeze-and-excitation networks, 2019. URL https://arxiv.org/abs/1709.01507.

Koluguri, N. R., Park, T., and Ginsburg, B. Titanet: Neural model for speaker representation with 1d depth-wise separable convolutions and global context, 2021. URL https://arxiv.org/abs/2110.04410.

Nagrani, A., Chung, J. S., and Zisserman, A. Voxceleb: A large-scale speaker identification dataset. In *Interspeech 2017*, interspeech$_2$017, $pp.$ 2616–2620.$ISCA$, $August$2017.10.21437/$interspeech$.2017 − 950.URL.

Okabe, K., Koshinaka, T., and Shinoda, K. Attentive statistics pooling for deep speaker embedding. In *Interspeech 2018*, interspeech$_2$018, $pp.$ 2252–2256.$ISCA$, 2018.10.21437/$interspeech$.2018 − 993.URL.

Panayotov, V., Chen, G., Povey, D., and Khudanpur, S. Librispeech: An asr corpus based on public domain audio books. In *2015 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 5206–5210, 2015. 10.1109/ICASSP.2015.7178964.

Park, D. S., Chan, W., Zhang, Y., Chiu, C.-C., Zoph, B., Cubuk, E. D., and Le, Q. V. Specaugment: A simple data augmentation method for automatic speech recognition. In *Interspeech 2019*, interspeech$_2$019, $pp.$ 2613–2617.$ISCA$, 2019.10.21437/$interspeech$.2019

Romero, A., Ballas, N., Kahou, S. E., Chassang, A., Gatta, C., and Bengio, Y. Fitnets: Hints for thin deep nets, 2015. URL https://arxiv.org/abs/1412.6550.

Snyder, D., Chen, G., and Povey, D. Musan: A music, speech, and noise corpus, 2015. URL https://arxiv.org/abs/1510.08484.

Snyder, D., Garcia-Romero, D., Sell, G., Povey, D., and Khudanpur, S. X-vectors: Robust dnn embeddings for speaker recognition. In *2018 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 5329–5333, 2018. 10.1109/ICASSP.2018.8461375.

Wang, F., Cheng, J., Liu, W., and Liu, H. Additive margin softmax for face verification. *IEEE Signal Processing Letters*, 25(7):926–930, 2018. ISSN 1558-2361. 10.1109/lsp.2018.2822810. URL http://dx.doi.org/10.1109/LSP.2018.2822810.

Xiang, X., Wang, S., Huang, H., Qian, Y., and Yu, K. Margin matters: Towards more discriminative deep neural network embeddings for speaker recognition, 2019. URL https://arxiv.org/abs/1906.07317.

Zhang, Y., Lv, Z., Wu, H., Zhang, S., Hu, P., Wu, Z., yi Lee, H., and Meng, H. Mfa-conformer: Multi-scale feature aggregation conformer for automatic speaker verification, 2022. URL https://arxiv.org/abs/2203.15249.